LoRA vs. Full Fine-Tuning — Multi-Seed Summary

Multilingual misinformation detection · English, Sepedi, isiZulu · Seeds 619, 436, 777

1. Setup

Three seeds (619, 436, 777) × two fine-tuning methods (full fine-tune, all parameters updated; LoRA with r = 64, α = 128) = six runs. All share one dataset (13,877 rows retained from 13,997), fixed 80/10/10 splits, and identical classical TF-IDF baselines. Selection / early-stopping metric: F1. All figures are on the untouched test split (n = 1,393).

Classical baselines are seed-invariant (identical every run): TF-IDF + Logistic Regression F1 = 0.807, Linear SVC = 0.822, Multinomial NB = 0.827 (the best classical). The comparison below covers the three fine-tuned transformers, where LoRA and full fine-tuning differ.

Table 1. Trainable parameters under LoRA (r = 64, α = 128).

Model	Total params	LoRA trainable	Trainable %
MzansiLM-125M	128,266,752	3,257,344	2.54%
AfroXLMR-base-76L	296,356,612	2,951,426	1.00%
AfroXLMR-large-76L	567,235,588	7,343,106	1.29%

2. Per-Seed Results (Test)

Each table: full vs. LoRA for the three transformers. Δ F1 = Full − LoRA (positive = full fine-tuning ahead).

2.1 Seed 619

Model	FULL FINE-TUNE			LORA			Δ F1
	Acc	F1	AUC	Acc	F1	AUC	
MzansiLM-125M	0.833	0.848	0.912	0.777	0.787	0.867	+0.061
AfroXLMR-base-76L	0.859	0.863	0.938	0.839	0.838	0.911	+0.025
AfroXLMR-large-76L	**0.881**	**0.889**	**0.956**	**0.885**	**0.887**	**0.946**	**+0.002**

2.2 Seed 436

Model	FULL FINE-TUNE			LORA			Δ F1
	Acc	F1	AUC	Acc	F1	AUC	
MzansiLM-125M	0.846	0.854	0.920	0.768	0.786	0.859	+0.068
AfroXLMR-base-76L	0.847	0.860	0.934	0.800	0.821	0.897	+0.039
AfroXLMR-large-76L	**0.901**	**0.903**	**0.963**	**0.858**	**0.865**	**0.929**	**+0.038**

2.3 Seed 777

Model	FULL FINE-TUNE			LORA			Δ F1
	Acc	F1	AUC	Acc	F1	AUC	
MzansiLM-125M	0.848	0.851	0.923	0.774	0.792	0.863	+0.059
AfroXLMR-base-76L	0.848	0.854	0.932	0.805	0.819	0.895	+0.035
AfroXLMR-large-76L	**0.902**	**0.906**	**0.963**	**0.889**	**0.893**	**0.945**	**+0.013**

3. Average Across Seeds

Table 2. Mean ± sample SD over the three seeds (test F1), with mean accuracy and AUC. Δ F1 = mean Full − mean LoRA.

Model	FULL FINE-TUNE			LORA			Δ F1
	Acc	F1 (mean ± sd)	AUC	Acc	F1 (mean ± sd)	AUC	
MzansiLM-125M	0.842	0.851 ± 0.003	0.918	0.773	0.788 ± 0.003	0.863	+0.063
AfroXLMR-base-76L	0.851	0.859 ± 0.005	0.935	0.815	0.826 ± 0.010	0.901	+0.033
AfroXLMR-large-76L	**0.895**	**0.899 ± 0.009**	**0.961**	**0.877**	**0.882 ± 0.015**	**0.940**	**+0.018**

Full fine-tuning leads on every model and every seed. The gap shrinks monotonically with scale — +0.063 F1 (MzansiLM-125M), +0.033 (AfroXLMR-base), +0.018 (AfroXLMR-large) — while LoRA trains 1–2.5% of parameters. LoRA is also the higher-variance regime at every scale (e.g. large: sd 0.015 vs 0.009).

4. AfroXLMR-large-76L — Focus (best model)

Table 3. AfroXLMR-large test F1 by seed and method.

Seed	Full F1	LoRA F1	Δ F1
619	0.889	0.887	+0.002
436	0.903	0.865	+0.038
777	0.906	0.893	+0.013
Mean ± sd	**0.899 ± 0.009**	**0.882 ± 0.015**	**+0.018**

The single-seed gap ranges from +0.002 (seed 619) to +0.038 (seed 436) — the spread within LoRA (0.865–0.893) exceeds the mean method gap, so a one-seed comparison is unreliable. Seed 619 alone would understate the difference; seed 436 alone would overstate it.

Table 4. AfroXLMR-large per-language test F1, mean over three seeds (African = Sepedi + isiZulu grouped).

Method	English	Sepedi	isiZulu	African
Full fine-tune	0.926	0.885	0.889	0.887
LoRA	0.910	0.865	0.870	0.867
Δ (Full − LoRA)	+0.016	+0.020	+0.019	+0.020

The full-vs-LoRA gap is fairly even across languages (+0.016 to +0.020), slightly wider on the lower-resource African languages than on English. Both methods keep the same ordering: English > isiZulu ≈ Sepedi.

5. Convergence — Winning Epoch and Losses

For each run, the winning epoch (highest validation F1, used for early stopping) and the training vs. validation loss at that epoch. The gap (Val − Train) is the overfitting signature: full fine-tuning drives training loss toward zero while validation loss climbs, opening a large gap; LoRA keeps the two close.

Table 5. Full fine-tune — winning epoch and loss at that epoch.

Model	Seed	Best epoch	Train loss	Val loss	Val − Train
MzansiLM-125M	619	3	0.0778	0.7815	+0.704
	436	4	0.0246	0.9870	+0.962
	777	2	0.3045	0.3270	+0.023
AfroXLMR-base-76L	619	5	0.1895	0.5508	+0.361
	436	4	0.2044	0.5728	+0.368
	777	6	0.1455	0.7242	+0.579
AfroXLMR-large-76L	619	2	0.2983	0.2643	−0.034
	436	5	0.0718	0.5593	+0.488
	777	5	0.1205	0.5221	+0.402

Table 6. LoRA (r = 64, α = 128) — winning epoch and loss at that epoch.

Model	Seed	Best epoch	Train loss	Val loss	Val − Train
MzansiLM-125M	619	5	0.3769	0.4087	+0.032
	436	5	0.3360	0.4272	+0.091
	777	6	0.2989	0.4279	+0.129
AfroXLMR-base-76L	619	8	0.3721	0.3705	−0.002
	436	5	0.3910	0.4199	+0.029
	777	4	0.4150	0.3760	−0.039
AfroXLMR-large-76L	619	10	0.2843	0.2774	−0.007
	436	3	0.3207	0.3282	+0.008
	777	8	0.2639	0.3020	+0.038

Mean winning epoch — full: MzansiLM 3.0, AfroXLMR-base 5.0, AfroXLMR-large 4.0; LoRA: MzansiLM 5.3, AfroXLMR-base 5.7, AfroXLMR-large 7.0. LoRA converges later and stops with a near-zero loss gap (−0.039 to +0.129), whereas full fine-tuning peaks early and stops already overfit — the gap reaches +0.96 on MzansiLM and +0.4 to +0.6 on the AfroXLMR models. Two full-FT large runs (seeds 436, 777) win at epoch 5 with training loss already near 0.07–0.12 and validation loss above 0.52.

6. Chi-Square Association (Best Model)

Pearson χ^2 test of independence between true and predicted label on each AfroXLMR-large test confusion matrix (2 × 2, df = 1, Yates-corrected). φ is the phi coefficient (= Cramér's V for a 2 × 2 table), an effect-size measure of association strength on 0–1.

Table 7. χ^2 for AfroXLMR-large, by method and seed (N = 1,393 each).

Method	Seed	χ^2 (df = 1)	p	φ
Full fine-tune	619	816.8	< 0.001	0.767
	436	893.2	< 0.001	0.802
	777	900.1	< 0.001	0.805
	mean	**870.0**	**—**	**0.791**
LoRA	619	823.4	< 0.001	0.770
	436	714.8	< 0.001	0.718
	777	891.0	< 0.001	0.801
	mean	**809.7**	**—**	**0.763**

Every run is significant far beyond p < 0.001 (exact p from $\approx 10^{-157}$ to 10^{-198}) — expected for a working classifier, so the test confirms rather than discriminates. The informative quantity is φ, which tracks F1: full fine-tuning averages φ = 0.791 vs. 0.763 for LoRA, and the single low value (0.718) is the weak seed-436 LoRA run. Chi-square of independence measures association within one model's predictions; it is not a test of Full vs. LoRA (that would need paired per-example predictions for a McNemar test).

7. Findings

- **Full fine-tuning wins consistently.** Across all three seeds and all three models (9 of 9 comparisons), full fine-tuning has the higher test F1. The direction of the result is unambiguous.
- **The margin is modest and scale-dependent.** Averaged over seeds, full leads by +0.063 F1 on MzansiLM-125M, +0.033 on AfroXLMR-base, and +0.018 on AfroXLMR-large. Bigger backbones close most of the gap.
- **Single-seed comparisons mislead.** On the large model the per-seed gap swings from +0.002 to +0.038; LoRA's own seed spread (0.028 F1) is larger than the mean method gap. Reporting one seed — as in the earliest comparison here — would have given a false read.
- **LoRA is the higher-variance regime** at every scale, so it benefits most from seed averaging.
- **LoRA regularizes; full fine-tuning overfits.** At the winning epoch, full fine-tuning already carries a large validation-minus-training loss gap (+0.4 to +0.96 on MzansiLM and AfroXLMR-base), while LoRA stops with the two losses near-equal (−0.04 to +0.13) and typically at a later epoch. LoRA trades a small F1 deficit for markedly more stable training.
- **Bottom line.** If maximum test F1 is the goal, full fine-tuning is reliably but modestly ahead (+0.018 on the best model). If parameter efficiency and training stability matter, LoRA recovers ~98% of full-FT F1 on AfroXLMR-large while training 1.29% of parameters — a favourable trade at scale, and a poor one on the small MzansiLM (−0.063).

8. Best Model — Full Results by Seed

AfroXLMR-large-76L (the retained model), test split, every seed and method. This is the headline results table for the study.

Table 8. AfroXLMR-large-76L test-set metrics by method and seed, with per-method means.

Method	Seed	Accuracy	Precision	Recall	F1	AUC
Full fine-tune	619	0.881	0.841	0.943	0.889	0.956
	436	0.901	0.890	0.918	0.903	0.963
	777	0.902	0.875	0.939	0.906	0.963
	mean	**0.895**	**0.869**	**0.933**	**0.899**	**0.961**
LoRA	619	0.885	0.885	0.888	0.887	0.946
	436	0.858	0.833	0.899	0.865	0.929
	777	0.889	0.869	0.918	0.893	0.945
	mean	**0.877**	**0.862**	**0.902**	**0.882**	**0.940**

Best single run overall: AfroXLMR-large, full fine-tune, seed 777 — F1 = 0.906, AUC = 0.963. Across methods, full fine-tuning trades precision for recall (mean recall 0.933 vs. LoRA 0.902; mean precision 0.869 vs. 0.862), so its F1 edge comes mainly from catching more misinformation.

9. Confusion Matrices (Best Model)

AfroXLMR-large-76L on the test split (N = 1,393), full fine-tune vs. LoRA for each seed. Rows = true class, columns = predicted; diagonal (correct) cells shaded. Order: non-misinformation, misinformation.

Seed 619

Full fine-tune — errors 166 / 1,393 (11.9%)

	Pred non-mis	Pred mis
True non-mis	**563**	126
True mis	40	**664**

LoRA — errors 160 / 1,393 (11.5%)

	Pred non-mis	Pred mis
True non-mis	**608**	81
True mis	79	**625**

Seed 436

Full fine-tune — errors 138 / 1,393 (9.9%)

	Pred non-mis	Pred mis
True non-mis	**609**	80
True mis	58	**646**

LoRA — errors 198 / 1,393 (14.2%)

	Pred non-mis	Pred mis
True non-mis	**562**	127
True mis	71	**633**

Seed 777

Full fine-tune — errors 137 / 1,393 (9.8%)

	Pred non-mis	Pred mis
True non-mis	**595**	94
True mis	43	**661**

LoRA — errors 155 / 1,393 (11.1%)

	Pred non-mis	Pred mis
True non-mis	**592**	97
True mis	58	**646**

The error split confirms the precision/recall trade: full fine-tuning's false negatives (misinformation missed) are lower on seeds 436 and 777 (58, 43) than LoRA's (71, 58), while false positives run comparable or lower. The one exception is seed 619, where LoRA has fewer false positives (81 vs 126) and edges full fine-tuning on total errors (160 vs 166) — the seed that produced the near-tie in F1.

10. Performance Ranking — Fine-Tuned vs. Classical

Best run per model type (highest test F1 across seeds) for each method, plus the three seed-invariant classical baselines. Ranked by F1 (the selection metric). Each metric column is shaded independently: green = best, red = worst in that column. Seed = the seed of the best run.

Table 9. Leaderboard — best model per type (Full / LoRA) against classical baselines, test split.

Rank	Model	Method	Seed	Acc	Prec	Recall	F1	AUC
1	AfroXLMR-large-76L	Full FT	777	0.902	0.875	0.939	0.906	0.963
2	AfroXLMR-large-76L	LoRA	777	0.889	0.869	0.918	0.893	0.945
3	AfroXLMR-base-76L	Full FT	619	0.859	0.851	0.875	0.863	0.938
4	MzansiLM-125M	Full FT	436	0.846	0.816	0.896	0.854	0.920
5	AfroXLMR-base-76L	LoRA	619	0.839	0.853	0.824	0.838	0.911
6	TF-IDF + Multinomial NB	classical	—	0.816	0.785	0.874	0.827	0.898
7	TF-IDF + Linear SVC	classical	—	0.814	0.795	0.851	0.822	0.899
8	TF-IDF + Logistic Reg.	classical	—	0.799	0.783	0.832	0.807	0.886
9	MzansiLM-125M	LoRA	777	0.774	0.740	0.852	0.792	0.863

AfroXLMR-large tops the board under both regimes (F1 0.906 full, 0.893 LoRA). The classical baselines cluster in the lower-middle (F1 0.807–0.827) — every AfroXLMR configuration beats them, but MzansiLM-125M under LoRA (F1 0.792) ranks last, the only fine-tuned setting that fails to beat TF-IDF. Full fine-tuning outranks its LoRA counterpart for all three model types.